Smart traffic lights that learn

A systematic literature review of reinforcement learning for traffic signal control: from simulation to real streets

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Why traffic lights should learn

Fixed-time light

Same green and red durations whether the road is empty or jammed

Learning light

Watches real traffic and changes its timing on the fly

Hundreds of studies report shorter waits **in simulation**, yet peer-reviewed tests on **real streets** are still very rare. Our review asks why this gap exists.

A smart system in a loop

01

Sense

Cameras and road sensors measure queues and waiting cars

02

Decide

An RL agent picks the next green phase

03

Act

The signal switches on the real road

04

Learn

Less waiting earns a reward, so the policy improves

> The light learns its timing by **trial and error**, not from a hand-made plan. Mistakes during learning are why real streets are hard.

Six research questions

Question	What we ask	What we extract
RQ1 Landscape	How fast is the field growing, and where?	Year, venue, country
RQ2 Methods	Which RL methods, states, actions and rewards?	Algorithm, agents, reward
RQ3 Validation	Tested only in simulators, or on real roads?	Simulator, data, realism level
RQ4 Performance	How much do waits, queues and travel times drop?	Values against baselines
RQ5 Barriers	What stops RL lights from reaching real cities?	Barriers and proposed fixes
RQ6 Future	Which gaps remain, and what should come next?	Gap matrix

How we review: PRISMA 2020

The international standard for systematic reviews, in six stages.

01

Plan

Protocol, questions and criteria fixed before searching

02

Search

Scopus, Web of Science and IEEE Xplore, 2019–2026

03

Screen

Two reviewers: title and abstract, then full text

04

Assess quality

Five criteria scored 0–2, maximum 10 points

05

Extract

One codebook; every number traced to its table or page

06

Synthesise

Taxonomy, statistics, gaps and the PRISMA flow diagram

Search strategy

CONCEPT A · METHOD

Reinforcement learning

Q-learning · deep Q-network · actor-critic · policy gradient · PPO

AND

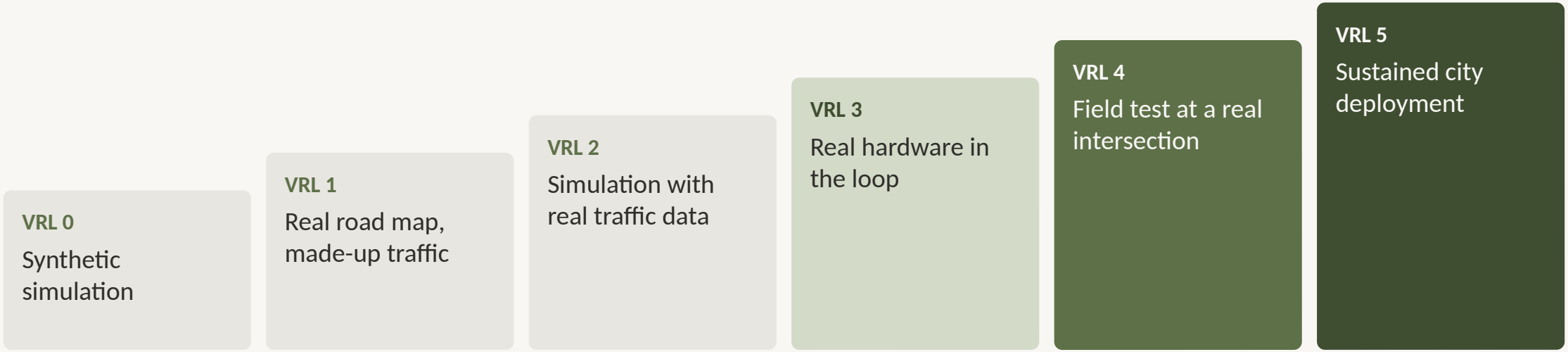
CONCEPT B · APPLICATION

Traffic signals

traffic light · signalized intersection · signal timing · signal phase

3	2019–2026	20
databases searched	publication window	known papers the pilot must find

Our key idea: a validation realism scale



Every study gets the highest level it reaches. Then we ask: **do the reported gains shrink as tests get more real?**

What the scoping search already shows

Simulators dominate

SUMO and CityFlow are the main test platforms in recent reviews

Field tests are rare

Deployment-focused papers from 2021–2022 report almost no live RL signals

The last SLR stops in 2020

Noaeen et al. (2022) reviewed 160 papers up to March 2020

Same data, many papers

Jinan, Hangzhou and New York datasets are reused again and again

Exploratory scoping search only: it shaped our design and is not counted as review evidence.

What we have built so far

15

planning and method documents, one 32-page dossier

3

database search strings, ready to run

64

extraction fields in one codebook

10

pipeline commands, from import to PRISMA diagram

> Every PRISMA count and statistic will be **computed from the records**, never typed by hand. All 10 automated tests pass.

Progress: about 50% complete

50%

The design half is done. The execution half starts with the database searches.

Phase	Weight	Status	Earned
Planning and protocol (Steps 1–6)	20%	Done	20
Search strategy (Steps 7–12)	15%	Pilot and lock left	13
Review tools (Steps 13–27)	15%	Built, tests pass	15
Database search and screening	20%	Next	0
Quality and data extraction	15%	Not started	0
Synthesis and writing	15%	Taxonomy drafted	2
Total	**100%**		**50**